

SOFTWARE CONFIGURATION

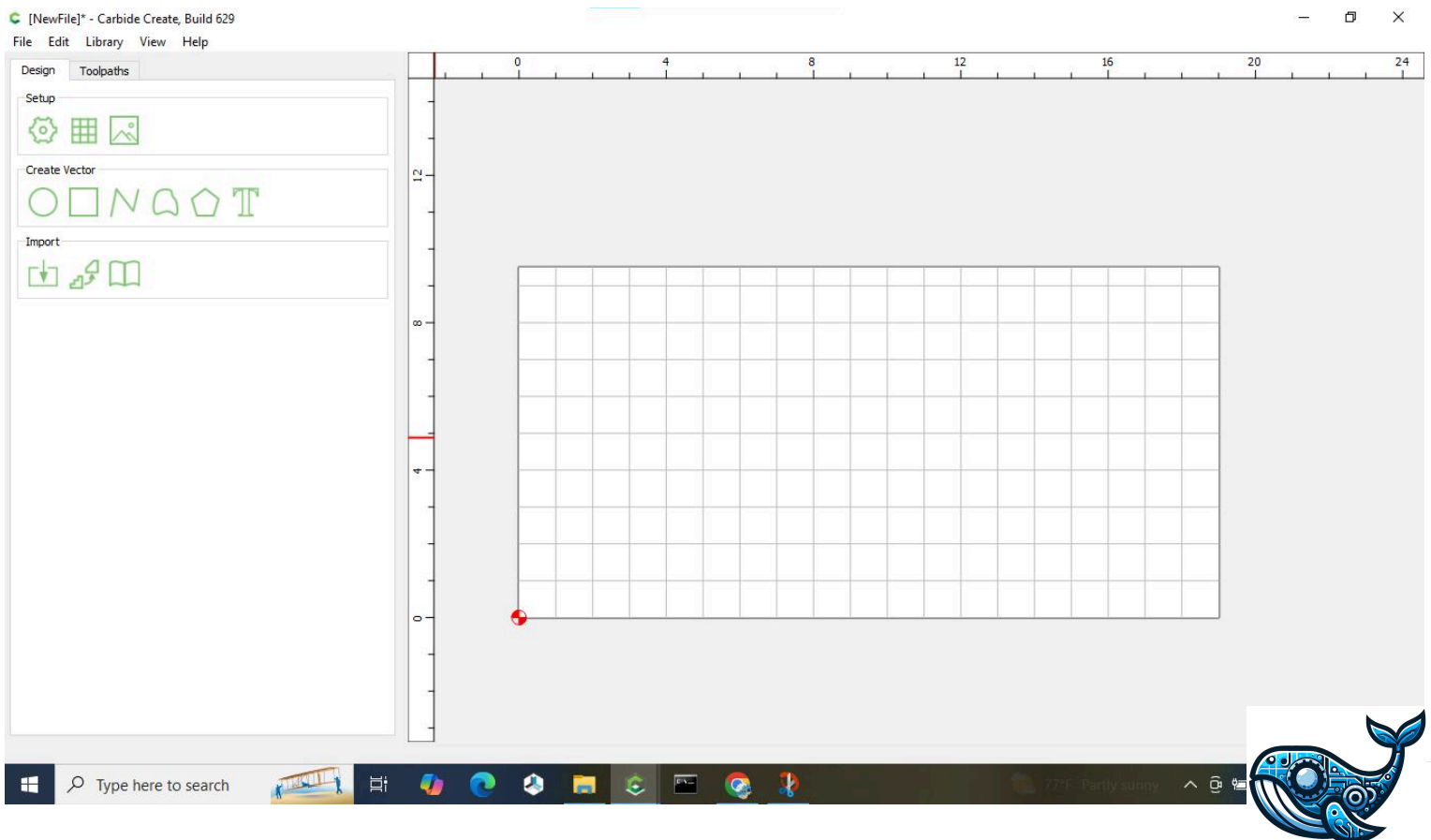
CARBIDE CREATE TUTORIAL

We use carbide create to generate the G-Code for our FRCNC.

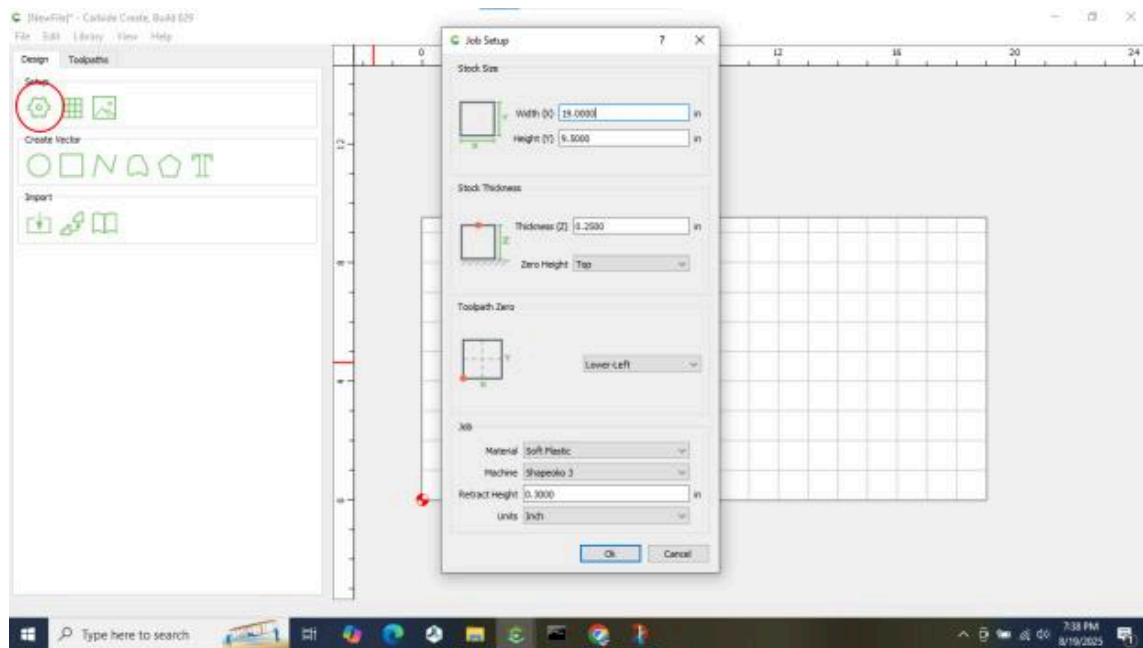
<https://carbide3d.com/blog/carbide-create-v6/>. The speeds/feeds noted here are for our V1 FRCNC. The V2 is capable of much faster material removal, but the instructions here are still applicable.

The next few pages lead you through a typical workflow to convert a DXF file (exported from a CAD program like Onshape) into a file that can be uploaded to FRCNC

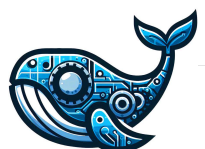
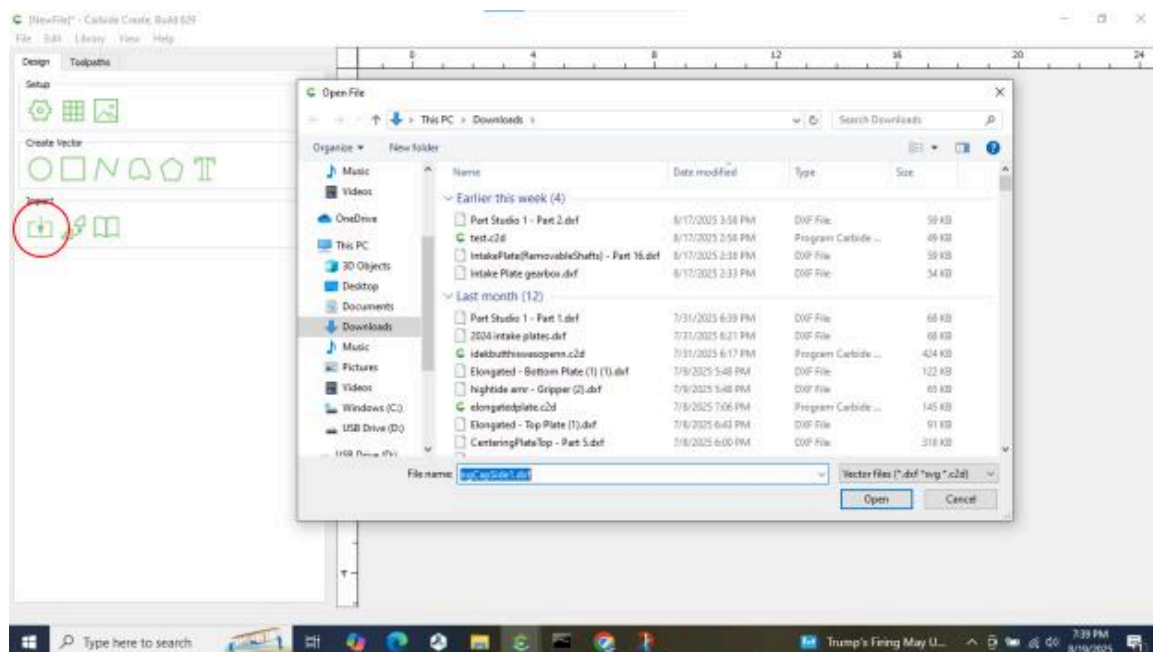
Note: Any tool that can generate g-code for GRBL can be used, and many alternatives exist (e.g. Fusion360, Onshape CAM Studio). So use whatever tools you are comfortable with!



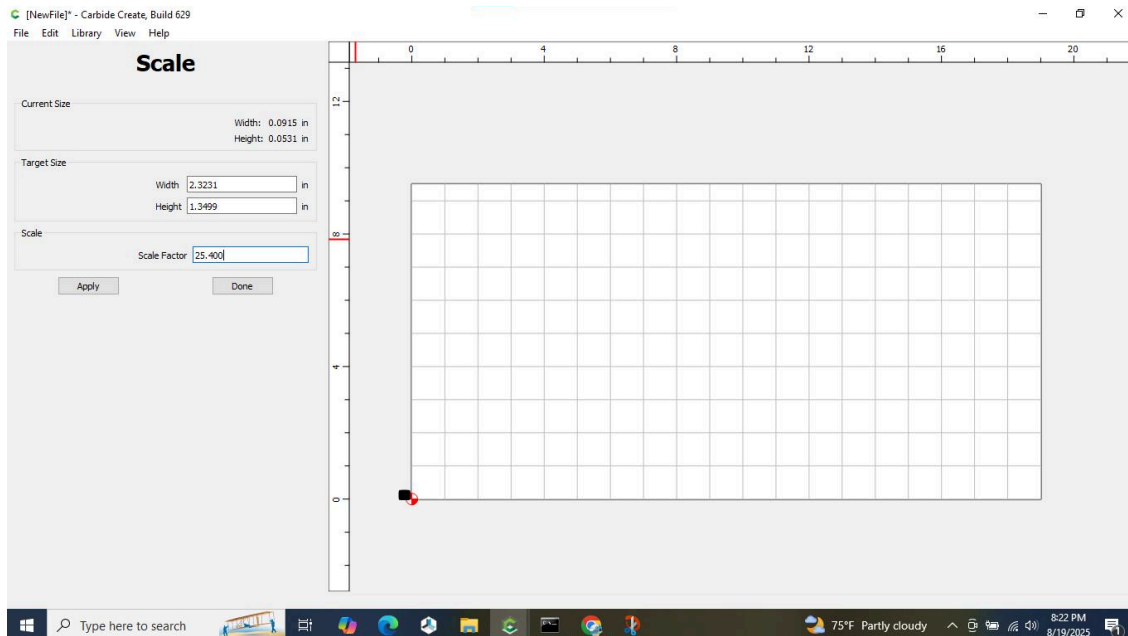
You can adjust the size of your cutting area by clicking the settings button under setup. You should change the Width (x) & Height (y) to the total width and height of the piece I am cutting. That way you can see how much space you have to work with and create toolpaths accordingly.



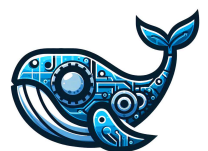
To import a DXF file, click the button with a box and an arrow right underneath the import section. Then select the file you wish to upload. The part I chose for this tutorial is a bearing cap/bearing plate.

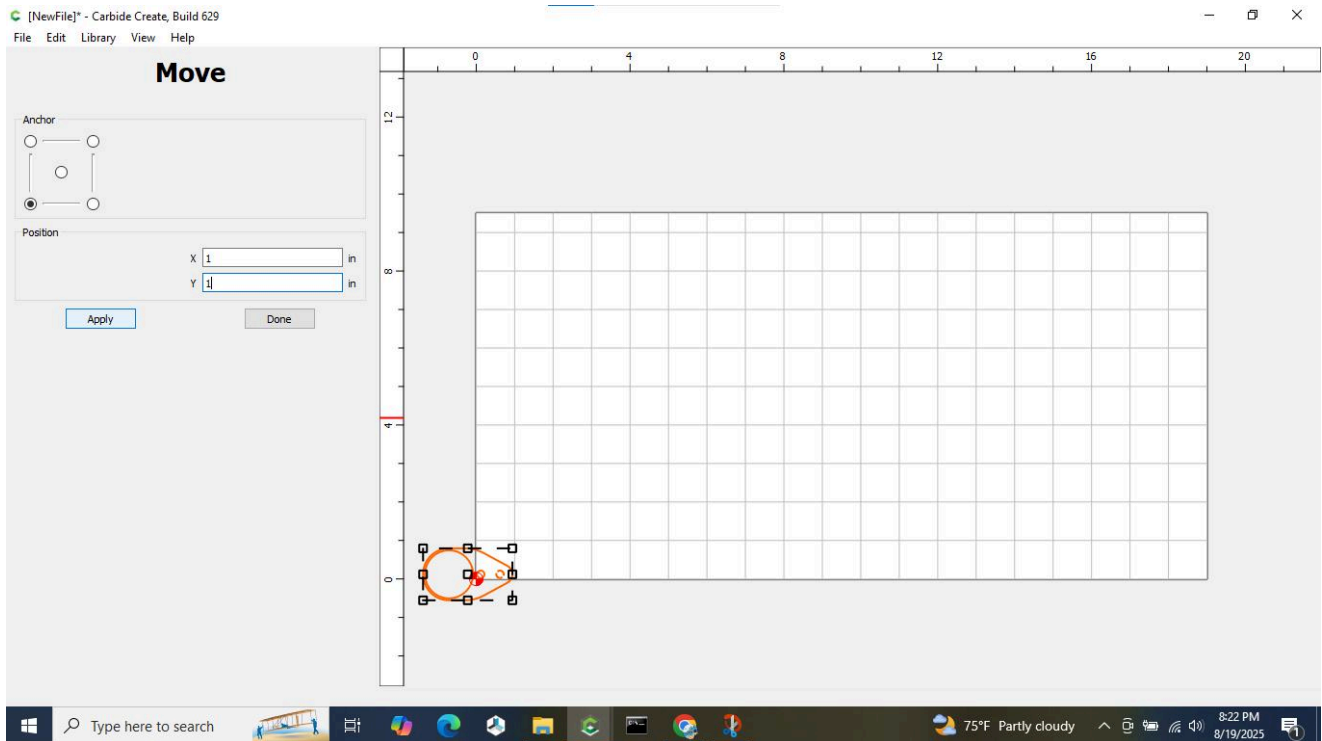


After you have selected the file you wish to cut, select it (it should be automatically selected if you just imported it) and click the scale button. Because you have exported the part from Onshape with release 13, you will sometimes need to scale any part you cut by 25.4 (mm to inches conversion factor) to bring it to its original size. Make sure to press the apply button.

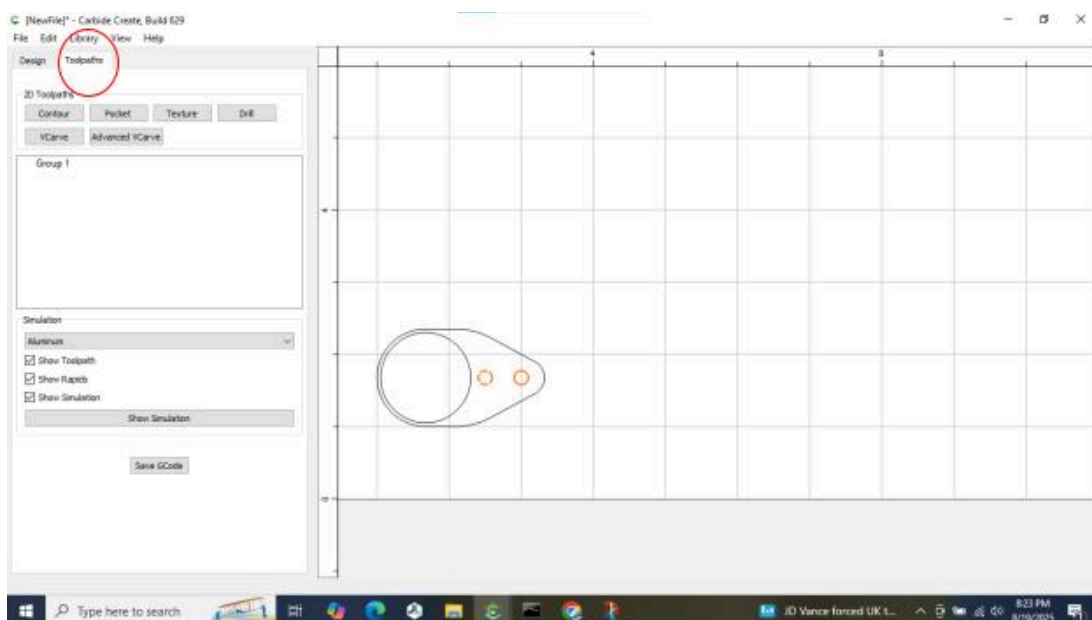


After you have scaled your part, you should be able to see your part. Then press the move button. In the “Move” panel, there is a grid shown. This grid has 4 corners and one midpoint, and describes what point the transformation will be applied to (ex. clicking on the lower left point, then moving to $X = 1$ and $Y = 1$ will make sure that the lower left corner of the part is located at $X = 1, Y = 1$). Usually, you will want to offset the lower left corner of the part to about (1, 1) or (.5, .5), that way, the toolpath will avoid mounting screws and so that imperfections in the edge of the stock will not affect the part. This, however, is just a general guideline, and is mainly up to your discretion and how big/small your piece is and where your mounting screws are. It should be completely in the grid given (image on next page).

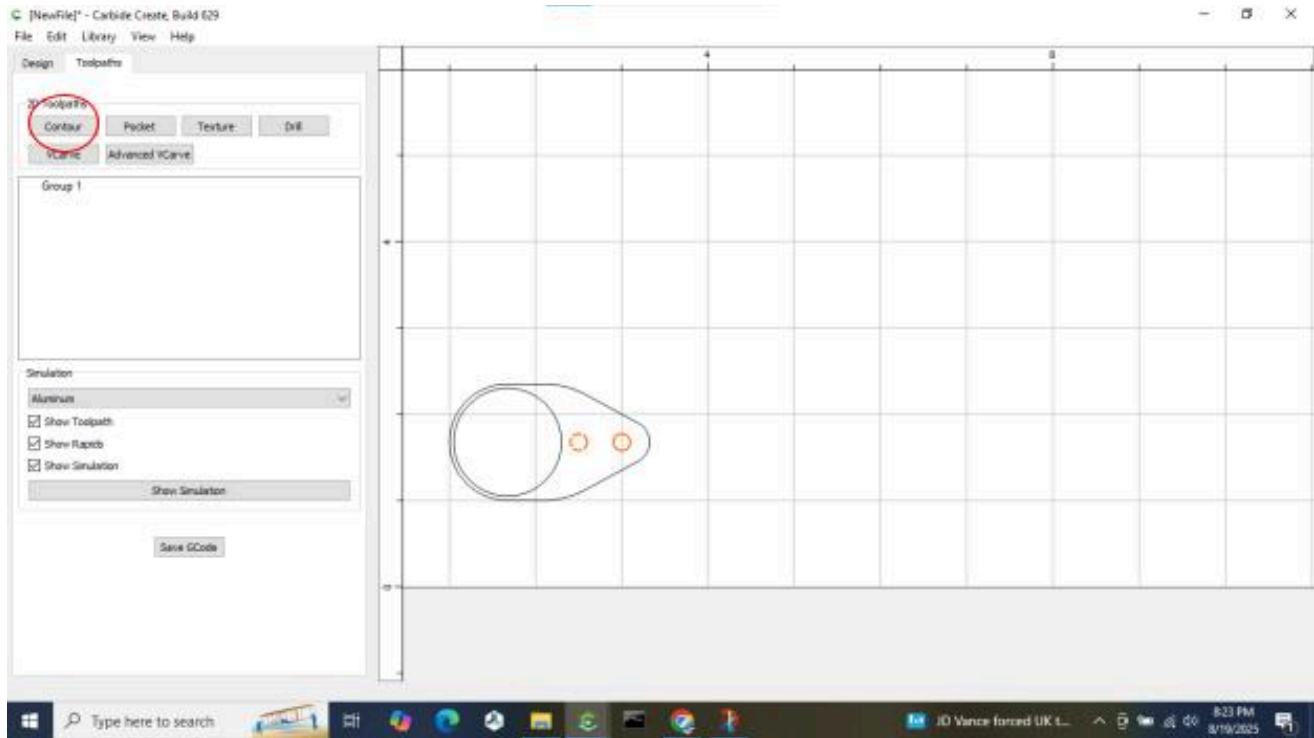




After you have moved your piece to the desired location, select **ONLY** the holes of the part (you must do this while still in the design page). When cutting it is important to cut holes first (in a separate code than the rest) so that you can put screws into these holes, and hold the part down, otherwise the part will go flying out when the full outline is cut out. Then go to the toolpaths page on the top left of the screen.

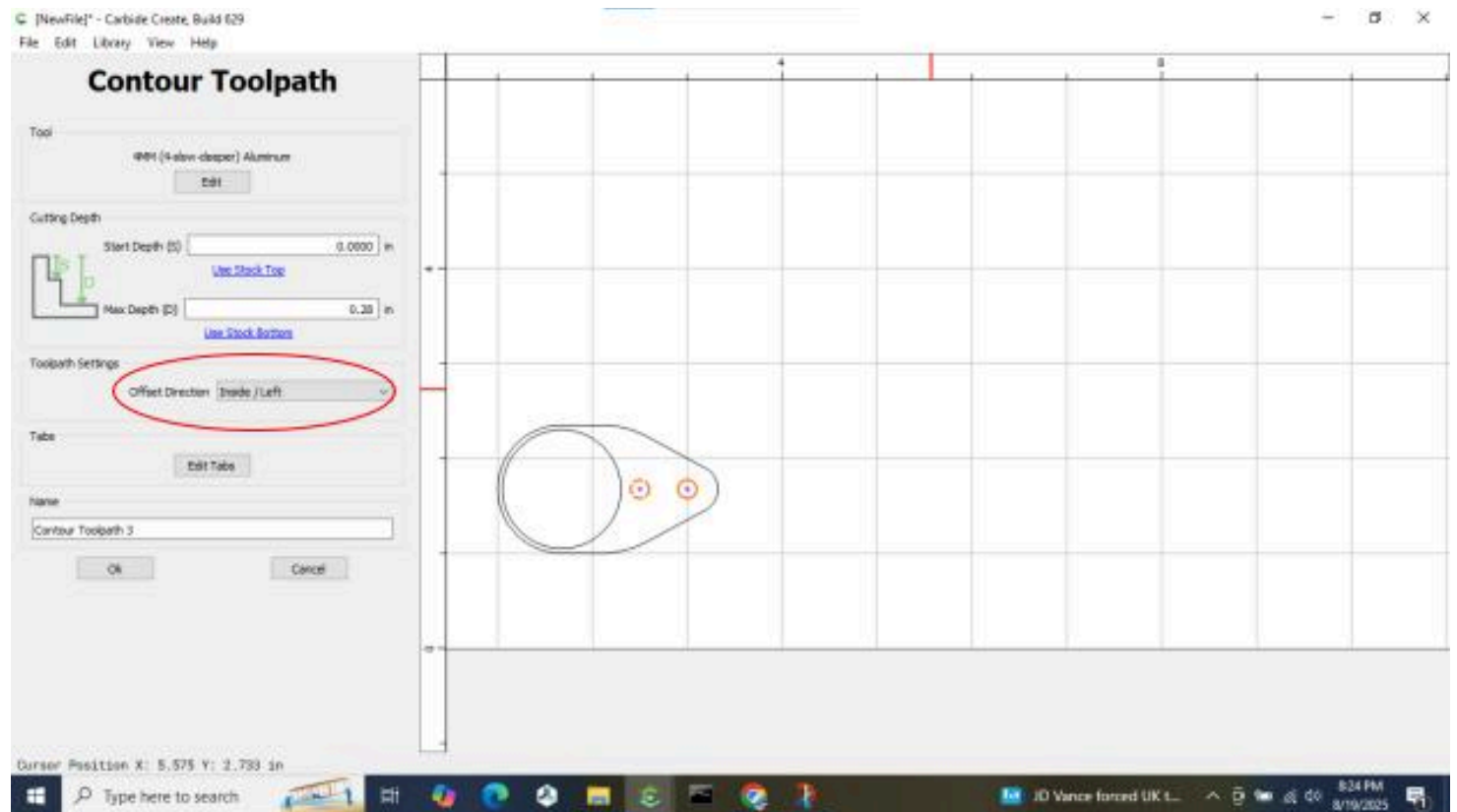


After you've gotten this far, select the toolpath option you want. Most of the time you will only be using contour (like shown below) but pocketing is very useful when you are creating trenches for cutting tubes. Right now focus on and select contour.

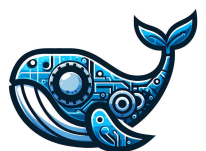
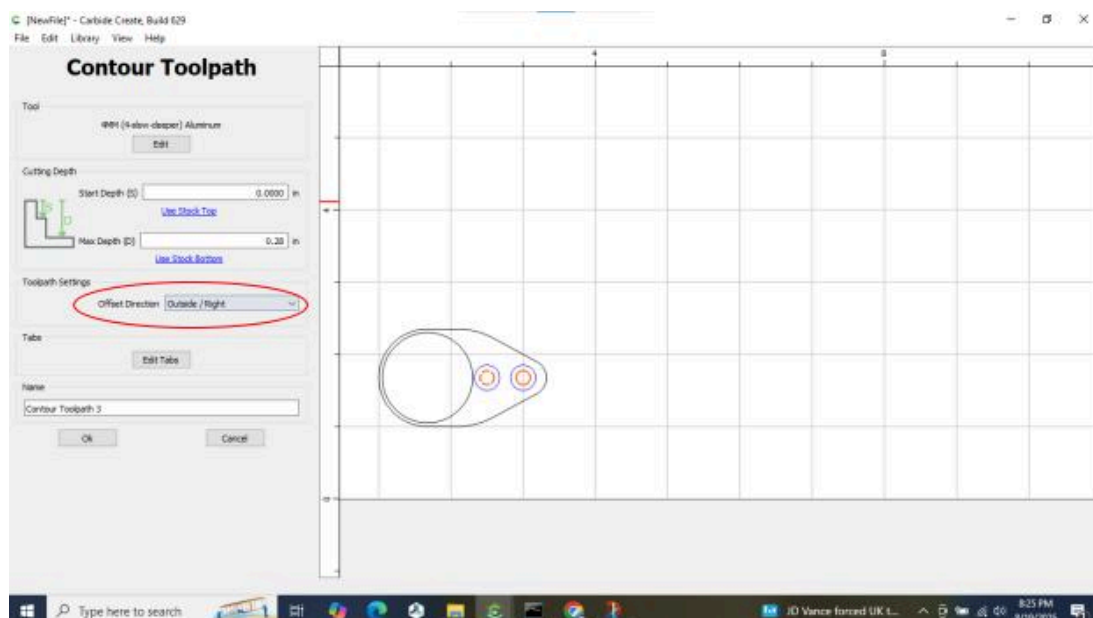


The contour toolpath page can look very overwhelming at first, so let's go step by step. First let's look at offset direction. Offset direction is essentially showing you how the spindle is going to drill a certain thing. The blue circle shows you the path that the spindle will take, and the orange circle shows you the outline that the toolpath is being traced around. For example, in the image on the next page, the offset direction is inside/left. This means the toolpath will be generated inside the selected geometry. For the offset direction outside/right, the toolpath will be generated tracing the outside of the selected geometry. In most cutting jobs, both of these offset directions will be used, in differing contexts

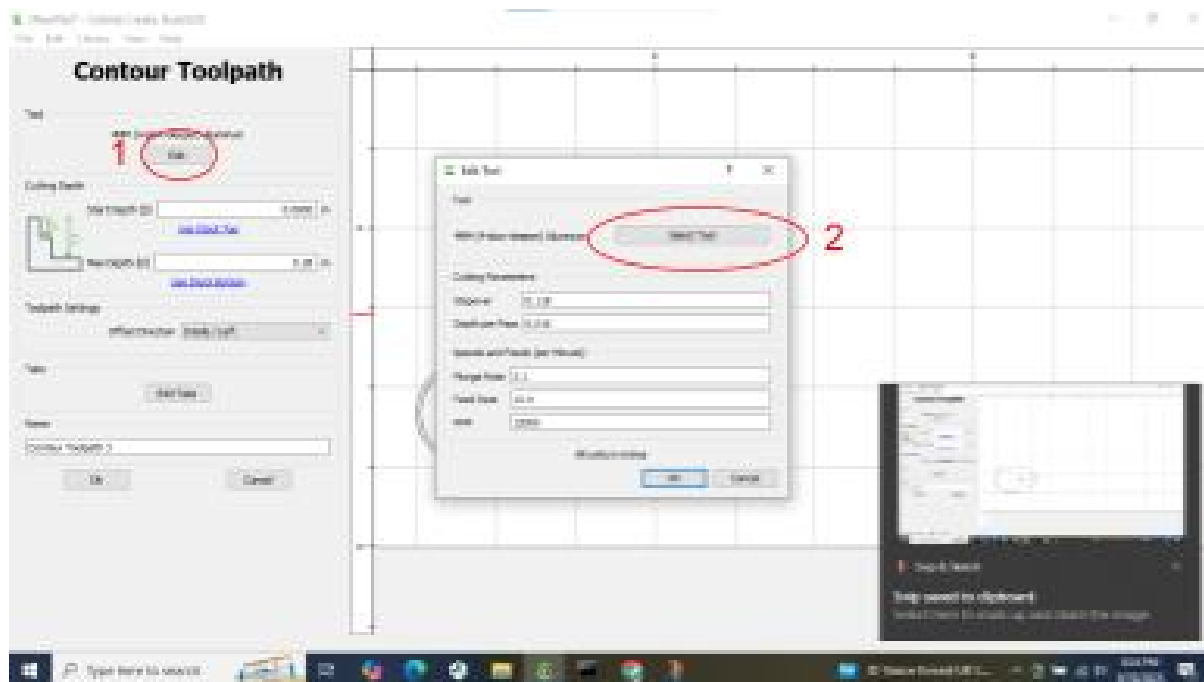




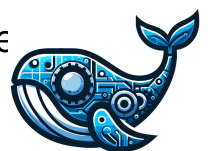
An example of the hole starting from the outside is below. If we see the offset direction, it is set as outside/right. This means the hole will begin cutting much bigger and get smaller until it reaches where the hole is on the toolpath. As you can imagine, it will make a much bigger hole than you originally wanted, as it will be cutting off all the material on the outside. I will explain why we would ever use outside/right as the instructions go on.



Go back to inside/left. Moving onto the tools category, push the edit button. This is where you can change all your cutting parameters, speeds & feeds, and the tool you will be using to cut the piece. Our settings for cutting parameters and speeds & feeds are shown below, feel free to copy them. The main thing you will be doing with this button is changing/selecting the tool you will be using for a specific piece. To change this push the select tool button.



After you've selected the select tool button, it will take you to this screen with lots of Carbide 3D files. Scroll all the way down till you get to the FRCNC ones. There are two main options (you will have to create these, numbers are on the next page); FRCNC-Any-Aluminum or FRCNC-Any-Hard Plastic. Select any aluminum if you are planning to cut metal, and any hard plastic if you are planning to cut wood/polycarb. For this instance I am cutting metal so we will click any aluminum. When you click it, it will take you to a drop down that says End Mills. Depending on the size of your bit select the 5mm option or the 4mm option. We mainly use 4mm bits. We have 2 aluminum tools for 4mm, just 4-slow and 4-slow-deeper. 4-slow-deeper seems to cut faster and more efficiently so we mainly use that one information for that tool is shown on the next page.



This profile is for polycarbonate and wood:

Select Tool

MachineAllMaterialAll

> Carbide 3D-Nomad Pro-Softwood

> Carbide 3D-Shapeoko HDM-Aluminum

> Carbide 3D-Shapeoko HDM-Hard Plastic

> Carbide 3D-Shapeoko HDM-Hardwood

> Carbide 3D-Shapeoko HDM-Soft Plastic

> Carbide 3D-Shapeoko HDM-Softwood

> Carbide 3D-Shapeoko-Aluminum

> Carbide 3D-Shapeoko-Hard Plastic

> Carbide 3D-Shapeoko-Hardwood

> Carbide 3D-Shapeoko-Soft Plastic

> Carbide 3D-Shapeoko-Softwood

> FRCNC-Any-Aluminum

> End Mills

> Ball Mills

> Vee

> Engravers

> FRCNC-Any-Hard Plastic

> End Mills

> 4MM Plastic

> 4.2MM drill

> 12.7

> 1/8MM Plastic

> 5MM Plastic

> Ball Mills

> Vee

> Engravers

New Library

Duplicate Library

Delete Library

Cutter Information

#10

Geometry

Diameter:4.000 mm

Flute Length:12.700 mm

Included Angle:0.0

Num Flutes:2

2D Cutting Parameters

Feed Rate:2000.0 mm/min

Cut Depth:1.000 mm

RPM:12000

3D Cutting Parameters

Feed Rate:2000.0 mm/min

Stepover:10.0 %

RPM:12000

Finish Allowance:2.540 mm

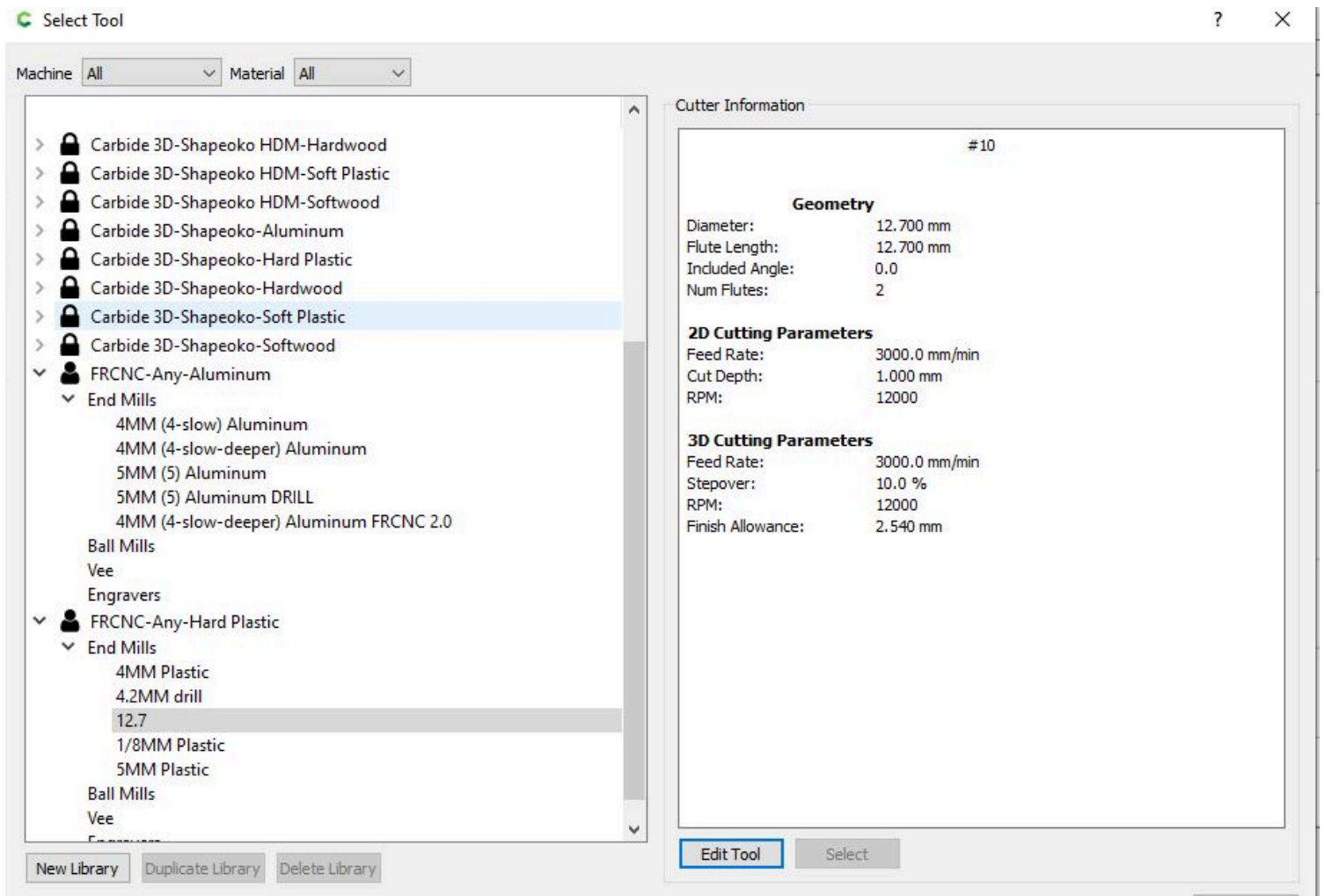
Edit Tool

Select

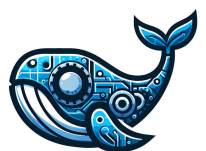
OK



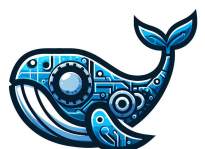
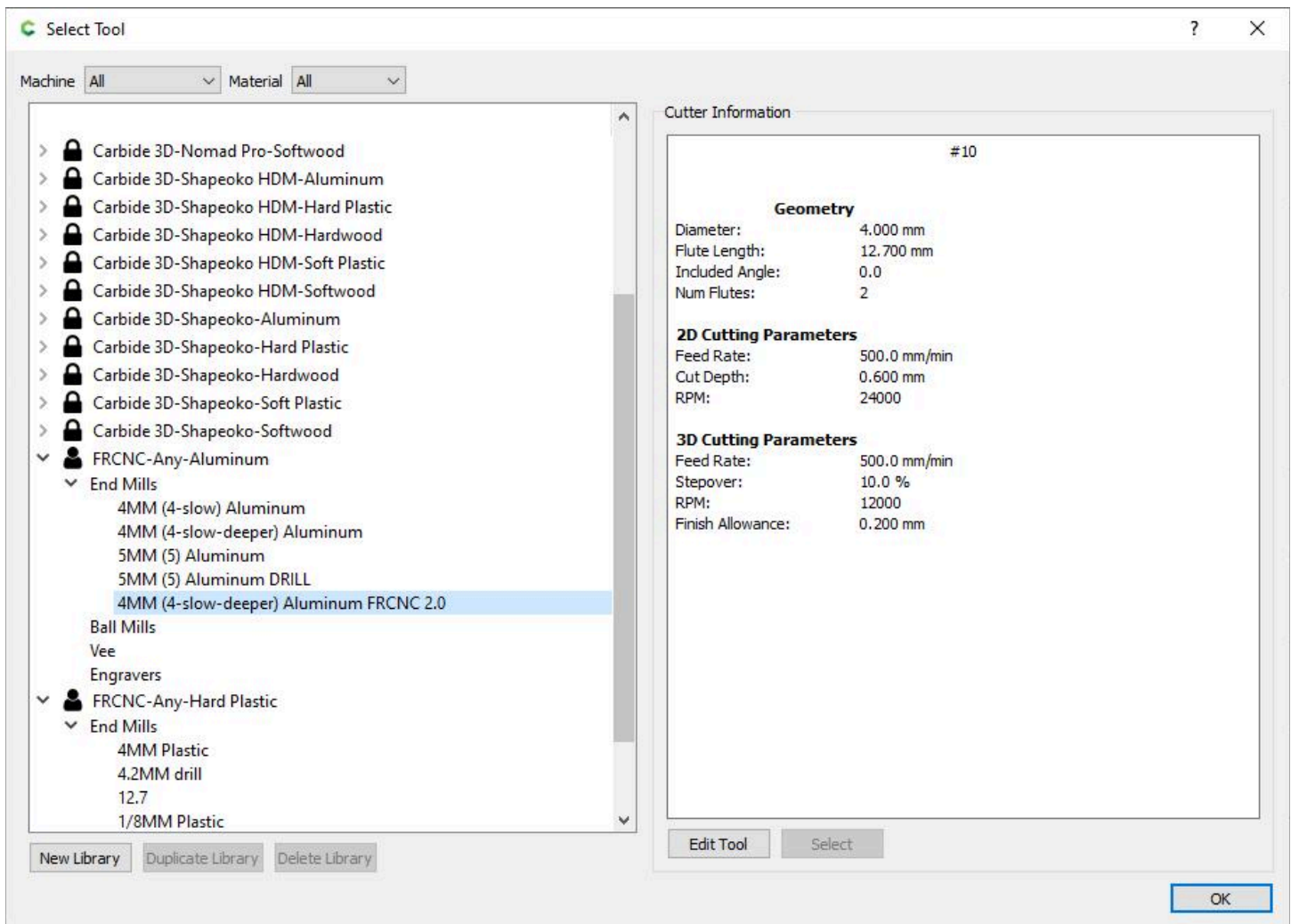
This profile is for standard plastics:



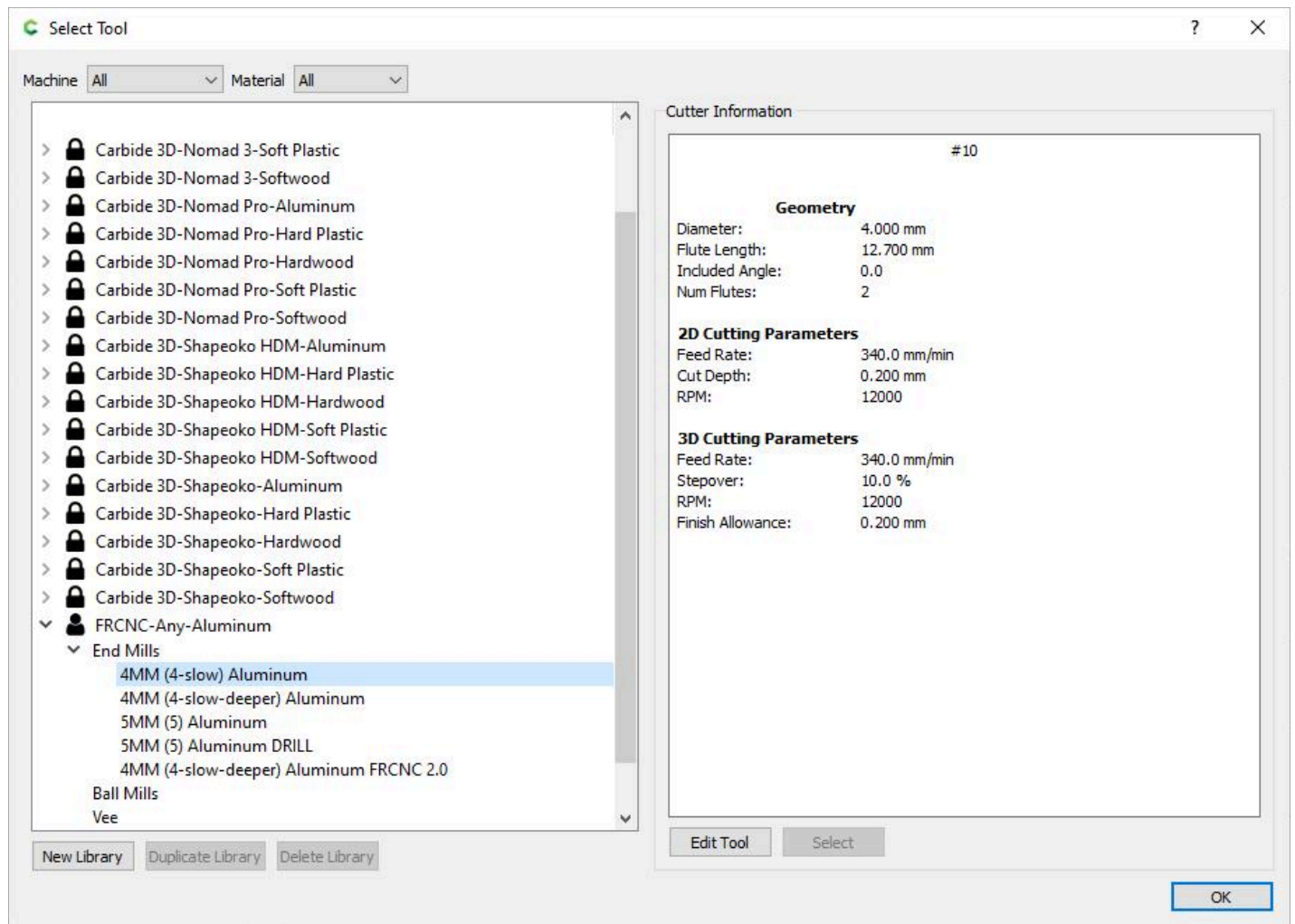
Note: 12.7 is for facing operations, stepover is 10% in the tool database. The actual stepover adjustment happens in the making of the toolpath, stepover can be set to 75%-90%



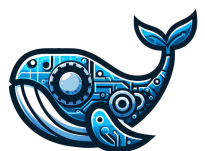
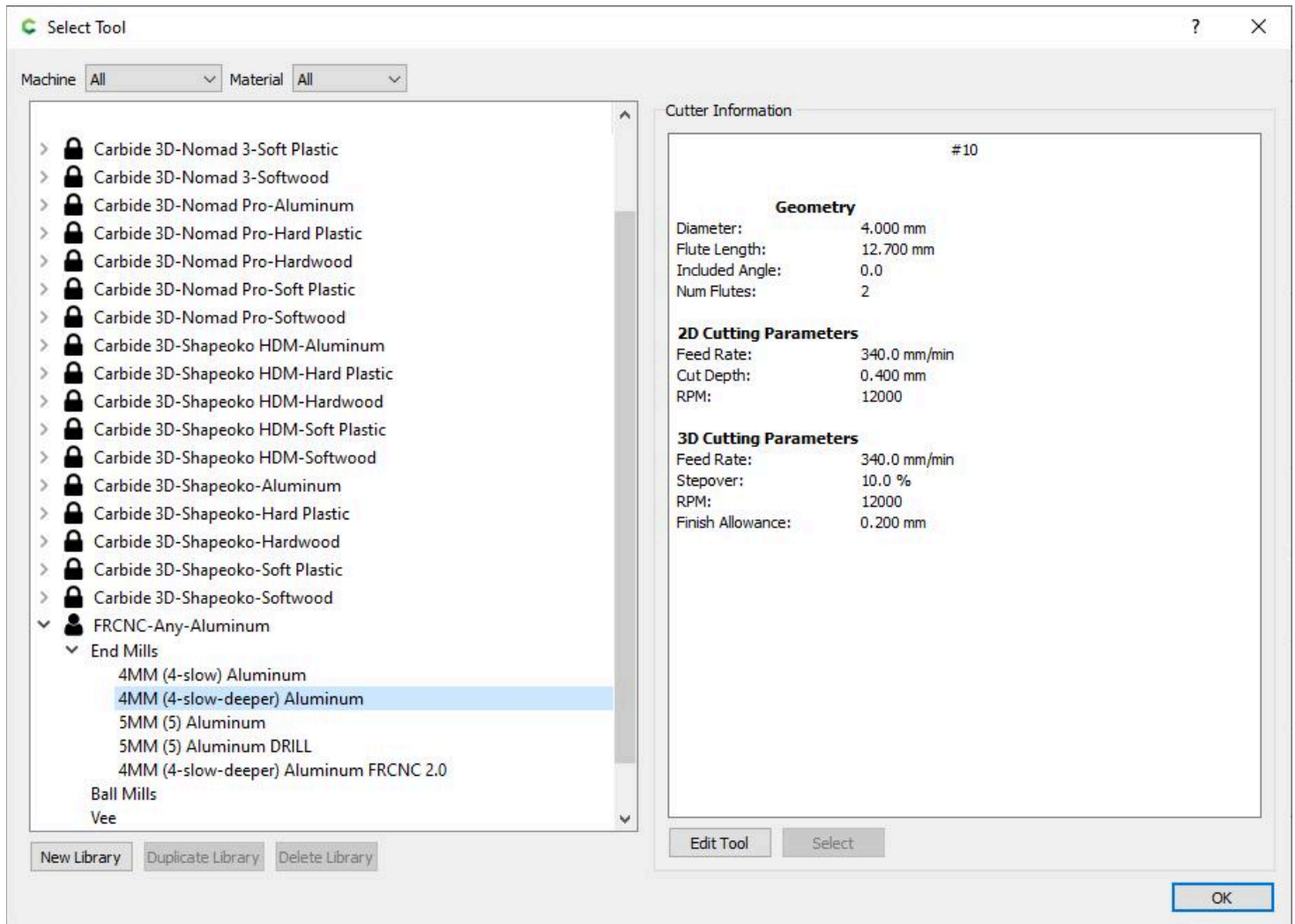
This tool profile is for the 1x6 or 2x6 aluminum tube FRCNC configuration. For FRCNC with extrusions, see next page.

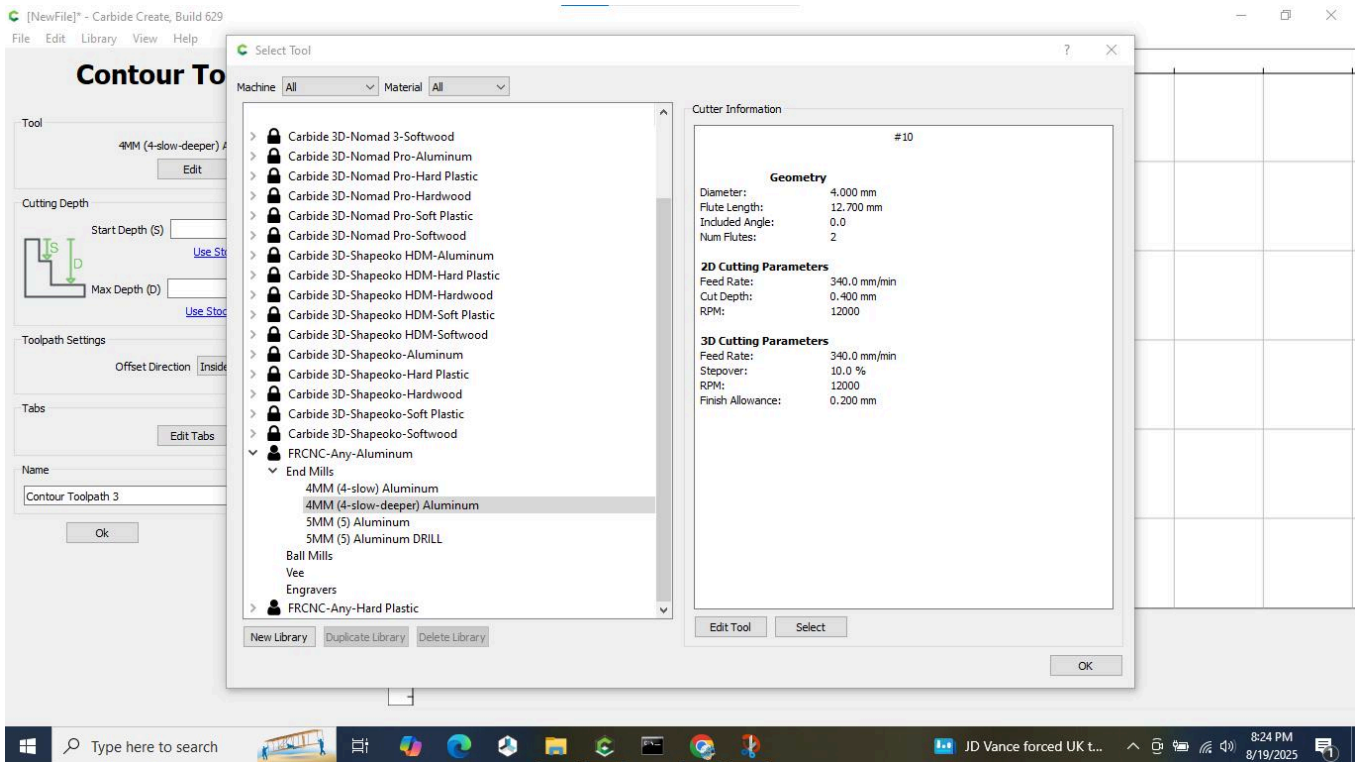


This tool profile is for V1 of the FRCNC. The preferred profile is slow-deeper (next page), but if you hear clattering, use slow.

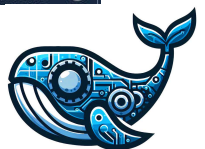
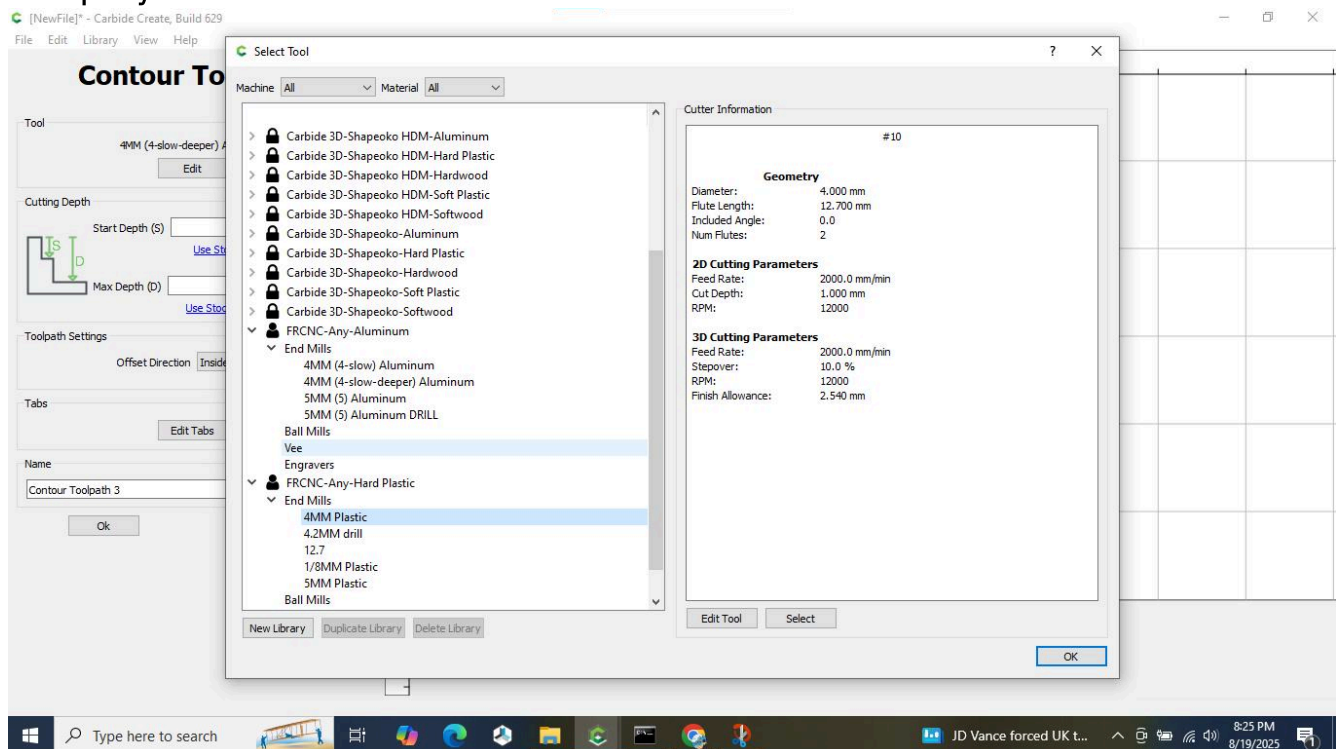


This tool profile is for V1 of the FRCNC, specifically the slow-deeper profile.

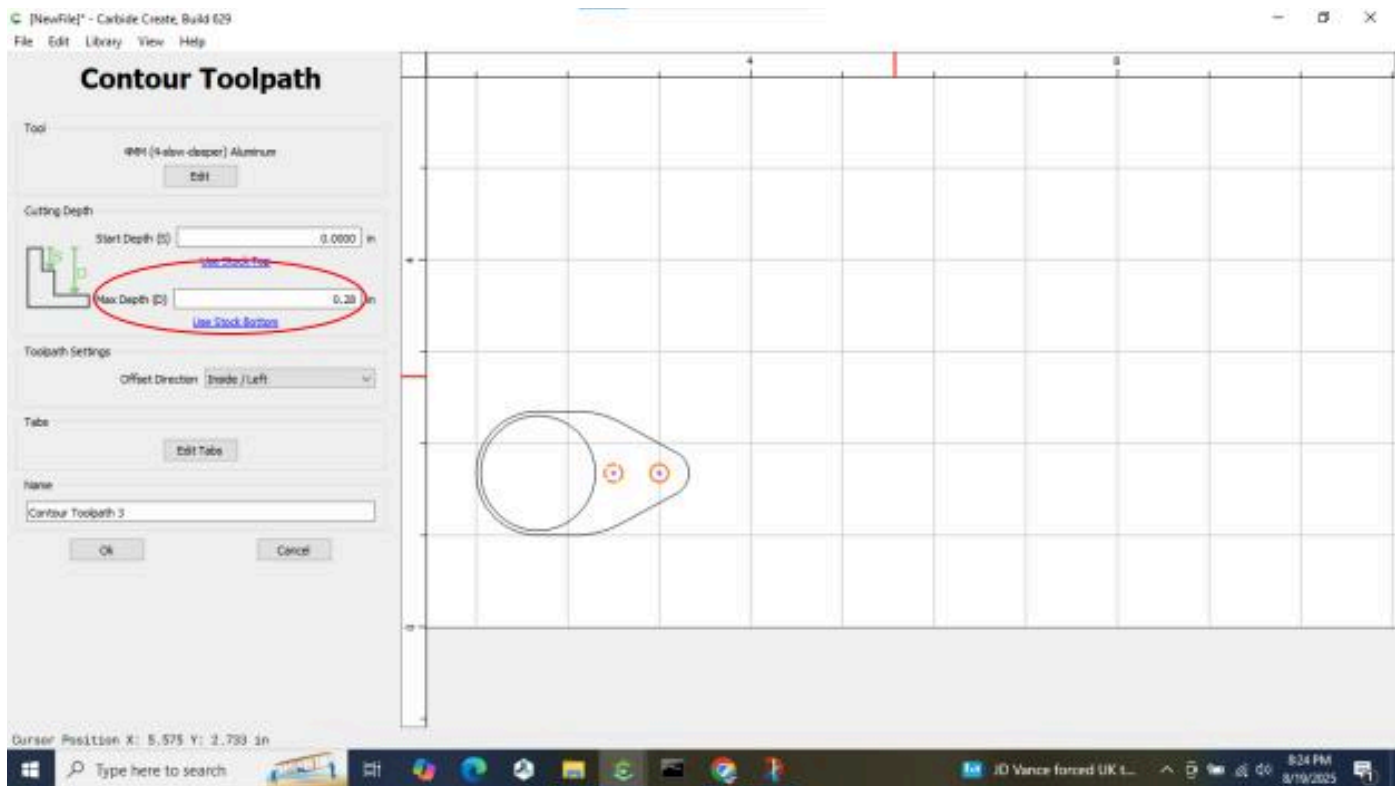




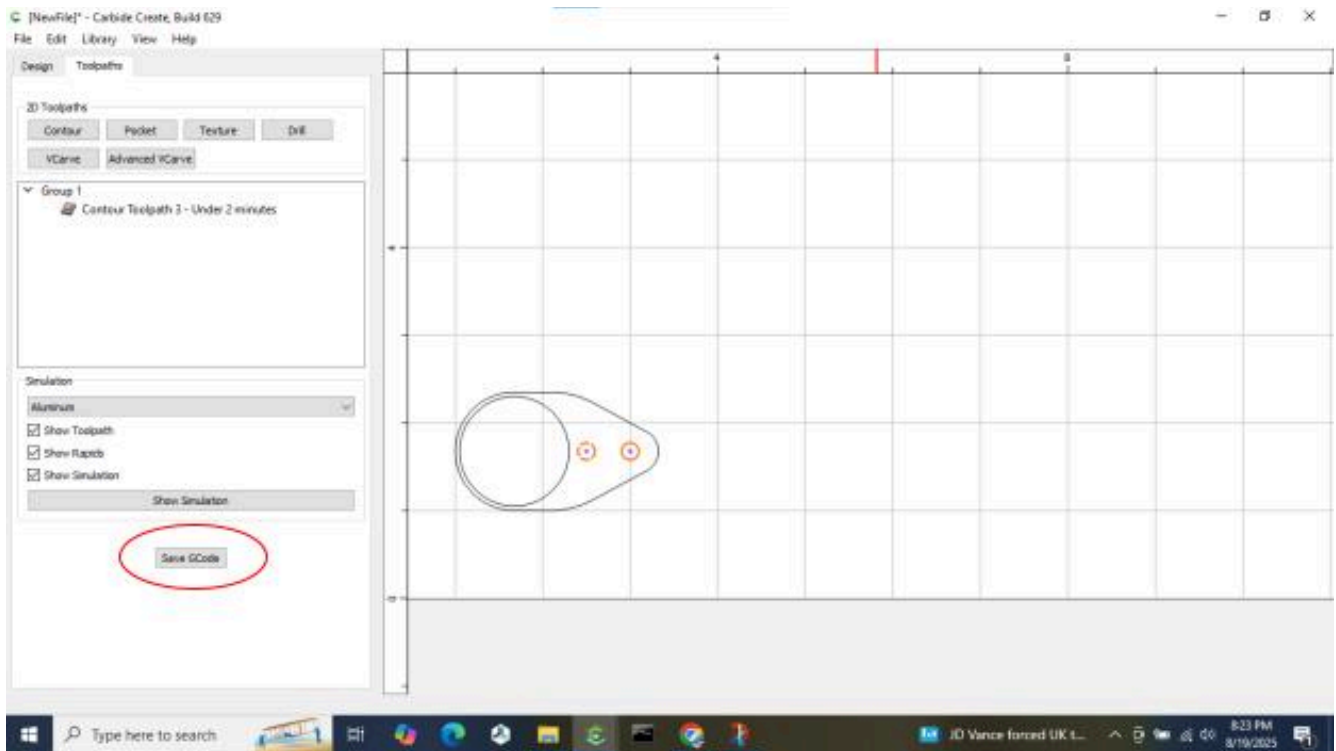
Let's say we were cutting polycarb. We would select the any-hard-plastic tool and choose the 4mm (again depending on your bit size you choose the tool you want) plastic. The information for that tool is also shown below. 4mm plastic can be used for both polycarbonate & wood.



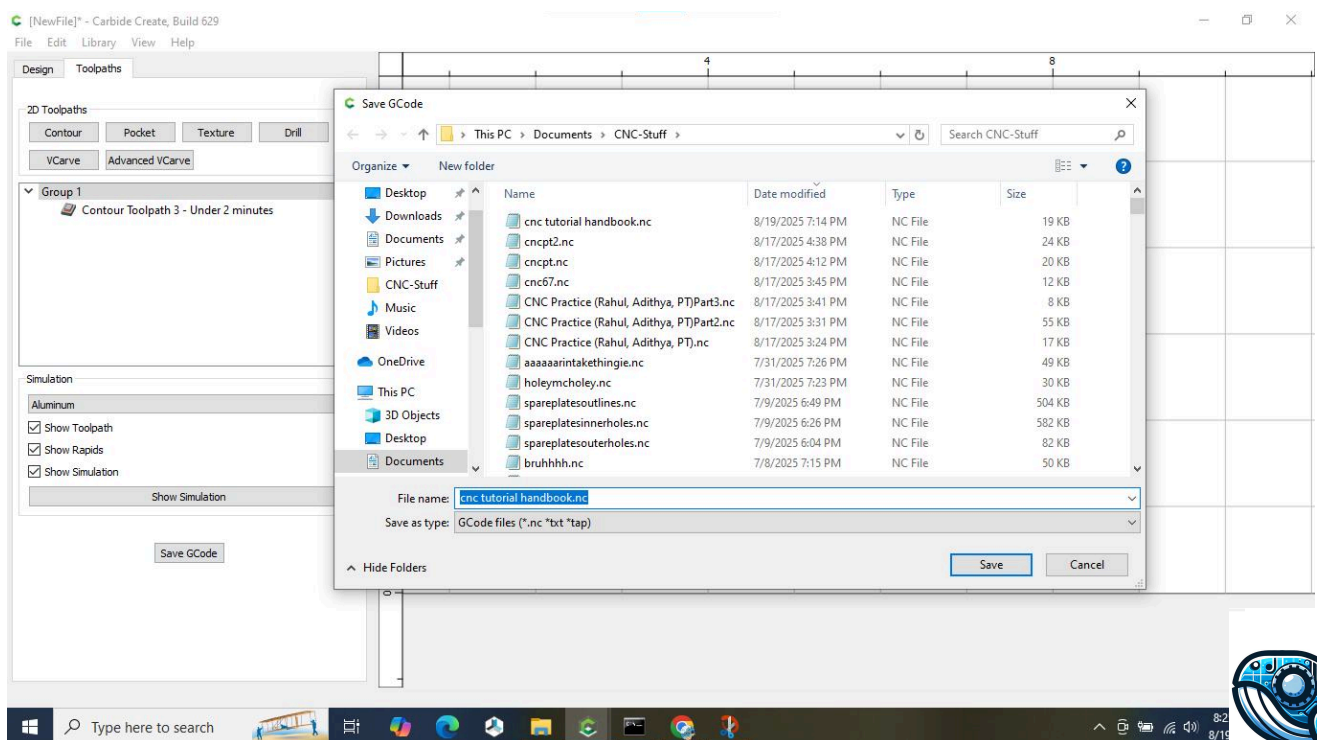
For this one though, we're using aluminum, so I selected the any-aluminum 4mm slow-deeper bit. Select the tool and then press OK. The last thing we will look at is called Max Depth. Max Depth is how deep you want the spindle to go. This depends on how thick the material you're cutting is. In this example I am using $\frac{1}{4}$ inch aluminum, which is 0.25 inches. Below I have put 0.28 as my max depth. This is because I find putting exactly the number of how thick the material is does not cut the material all the way through. To make up for that I usually add a good amount more to ensure it cuts all the way through (around 0.02 more then whatever the original was). Some examples would be if I was cutting $\frac{1}{8}$ polycarb I would probably do 0.14 instead of 0.125 just to make sure it goes all the way through.



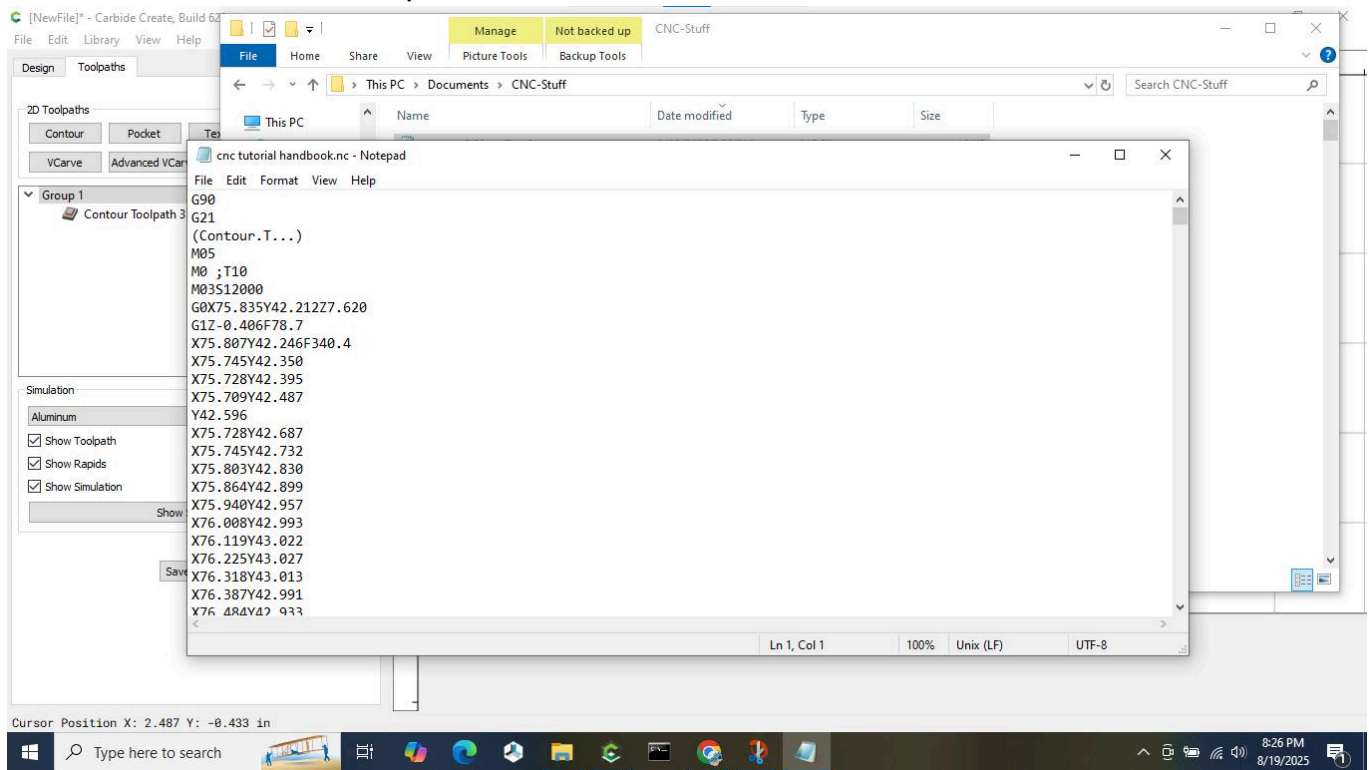
After you've chosen your max depth, press OK. Now you have the contour of your holes. From here you're going to want to export your GCode so you can put it on FluidNC. Press the save GCode button.



Once you've pressed that, save it as something you will remember. Make sure it's ending with .nc.

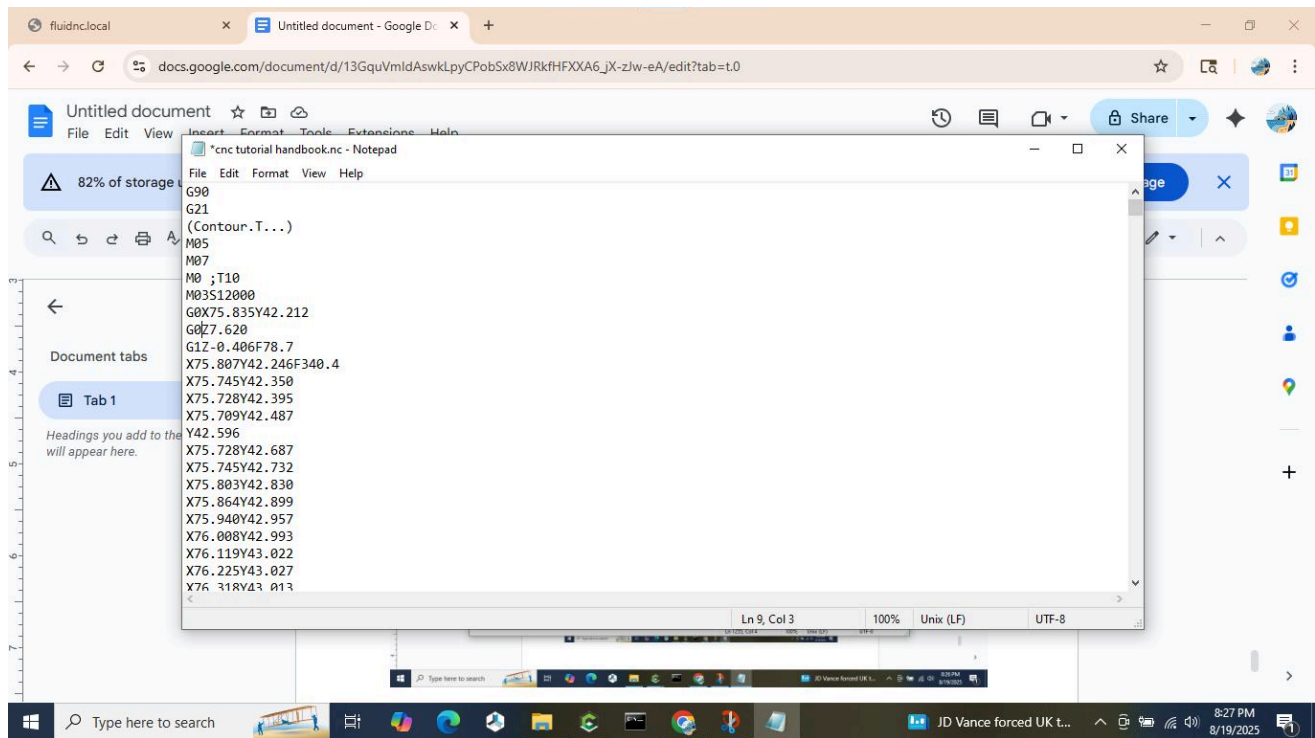


Then open the file, it will open a bunch of gcode, but you don't have to fuss on anything other than the first G0 line. This line is stating where the CNC will go when the code starts. So in this piece the CNC will start at X75.835, Y42,212, & Z7.620



Now that you've located this line, you're going to want to go right in front of where it says the Z coordinate and press enter. Then in that new line with just the Z add a G0 in the beginning which is telling the CNC to make sure to go there. The reason we do this is because the original line is telling the CNC to move to that x,y,z coordinate all at the same time. If we move to that area as we're matching the z value (or going down) the spindle will start to drag across the material and damage it.

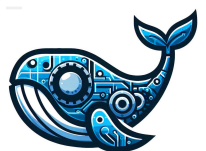




If you are doing aluminum, this step is **very important**. When cutting aluminum you need to make sure you have lubricant. Alcohol should be flowing at all times otherwise the spindle bit will melt and damage your piece as well as the bit. To stop this from happening, at the very beginning of your gcode where it says M05, press enter and write M07, this will ensure that whatever you are using for your mist setting will turn on at the beginning of the code, before the toolpath starts. (We changed this the M08 in our most recent configuration. You can add M07 or M08 depending on whichever configuration enables coolant/mist).

To stop the mist from coming out once the gcode/toolpath has ended, go to the last line in the gcode, and once again press enter after the M05. Add an M09 in the new line, which will stop mist from coming out completely.

Once you've done this, you can move on to working on the actual outline of your piece. You can delete or disable the holes toolpath you made by right clicking (I usually disable it, so it's easier if I ever need to go back to it).



Design Toolpaths

2D Toolpaths

Contour Pocket Texture Drill

VCarve Advanced VCarve

Group 1

Contour Toolpath 3 - Under 2 minutes

Edit Operation

Delete

Disable

Duplicate Toolpath

Simulation

Aluminum

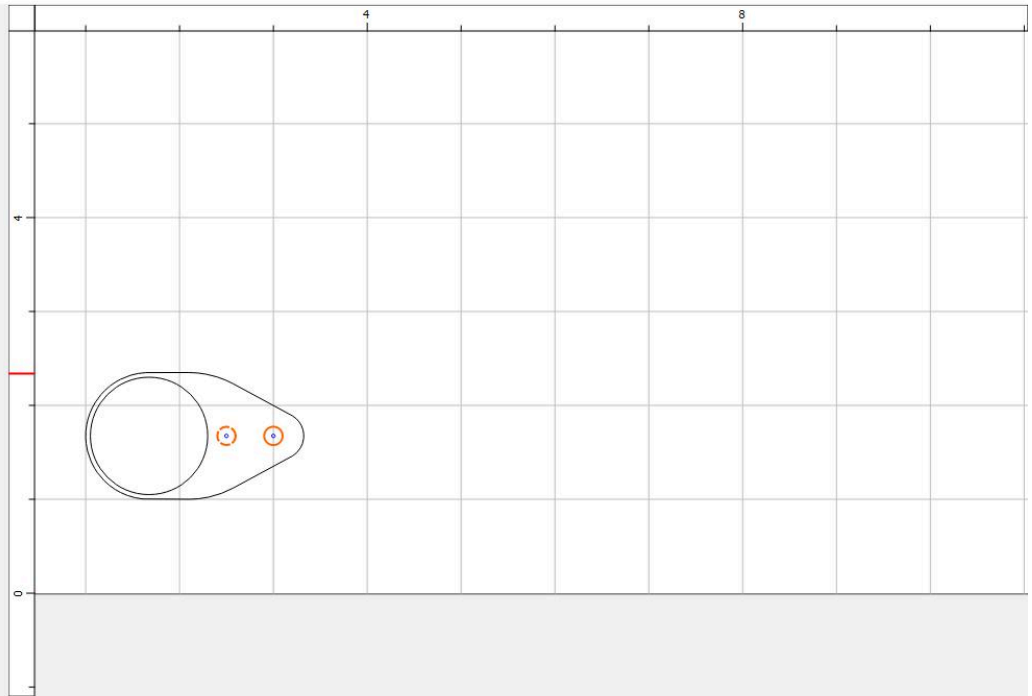
☒ Show Toolpath

☒ Show Rapids

☒ Show Simulation

Show Simulation

Save GCode



Then go back to the design page and select the outline of what you're cutting, and then go back to the toolpaths page.

Design Toolpaths

2D Toolpaths

Contour Pocket Texture Drill

VCarve Advanced VCarve

Group 1

Contour Toolpath 3 - Under 2 minutes

Simulation

Aluminum

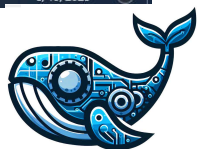
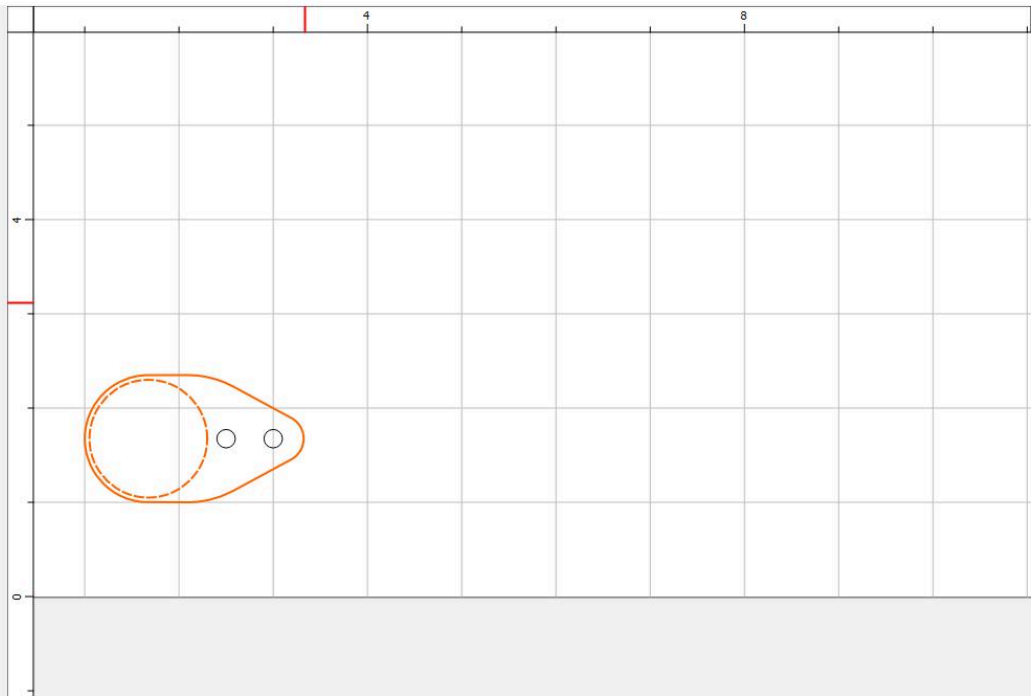
☒ Show Toolpath

☒ Show Rapids

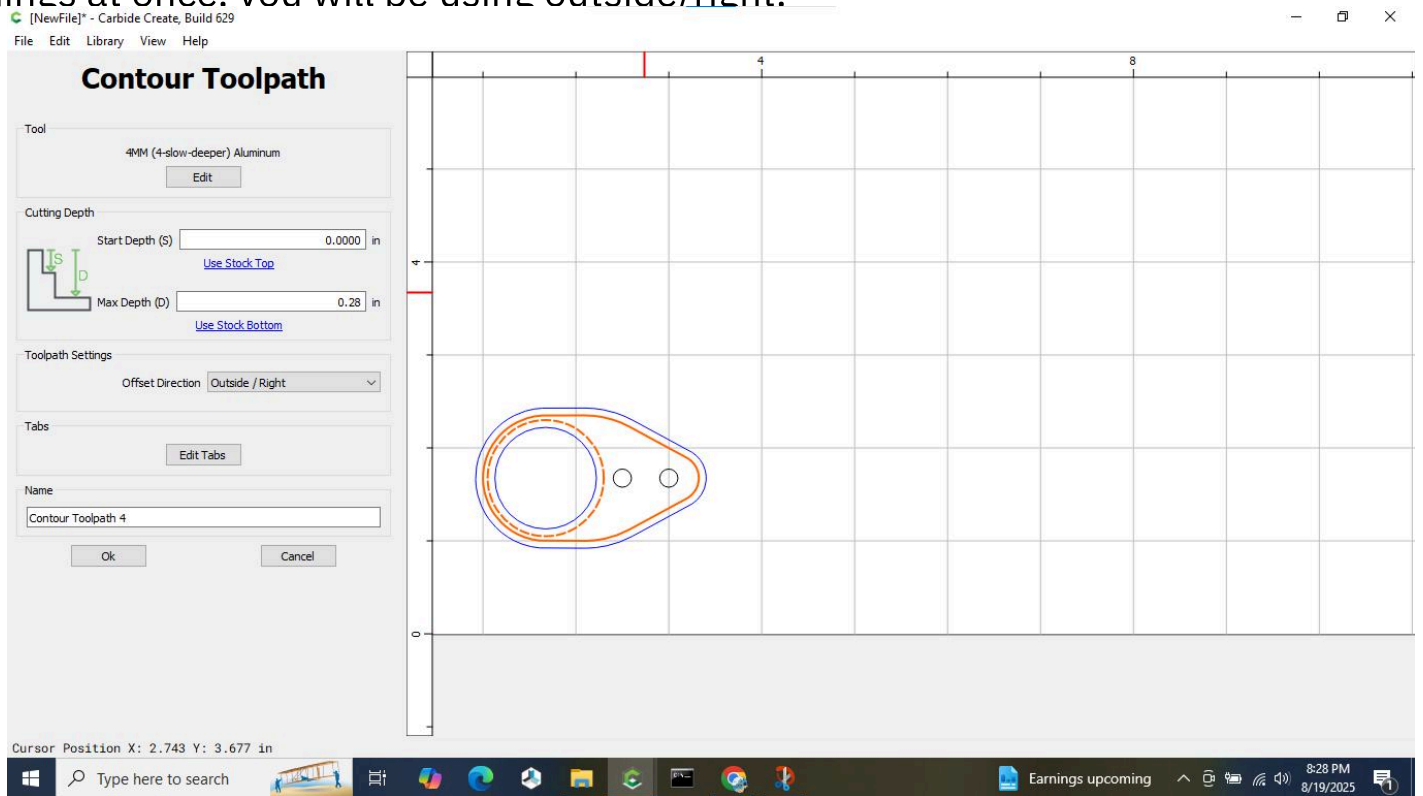
☒ Show Simulation

Show Simulation

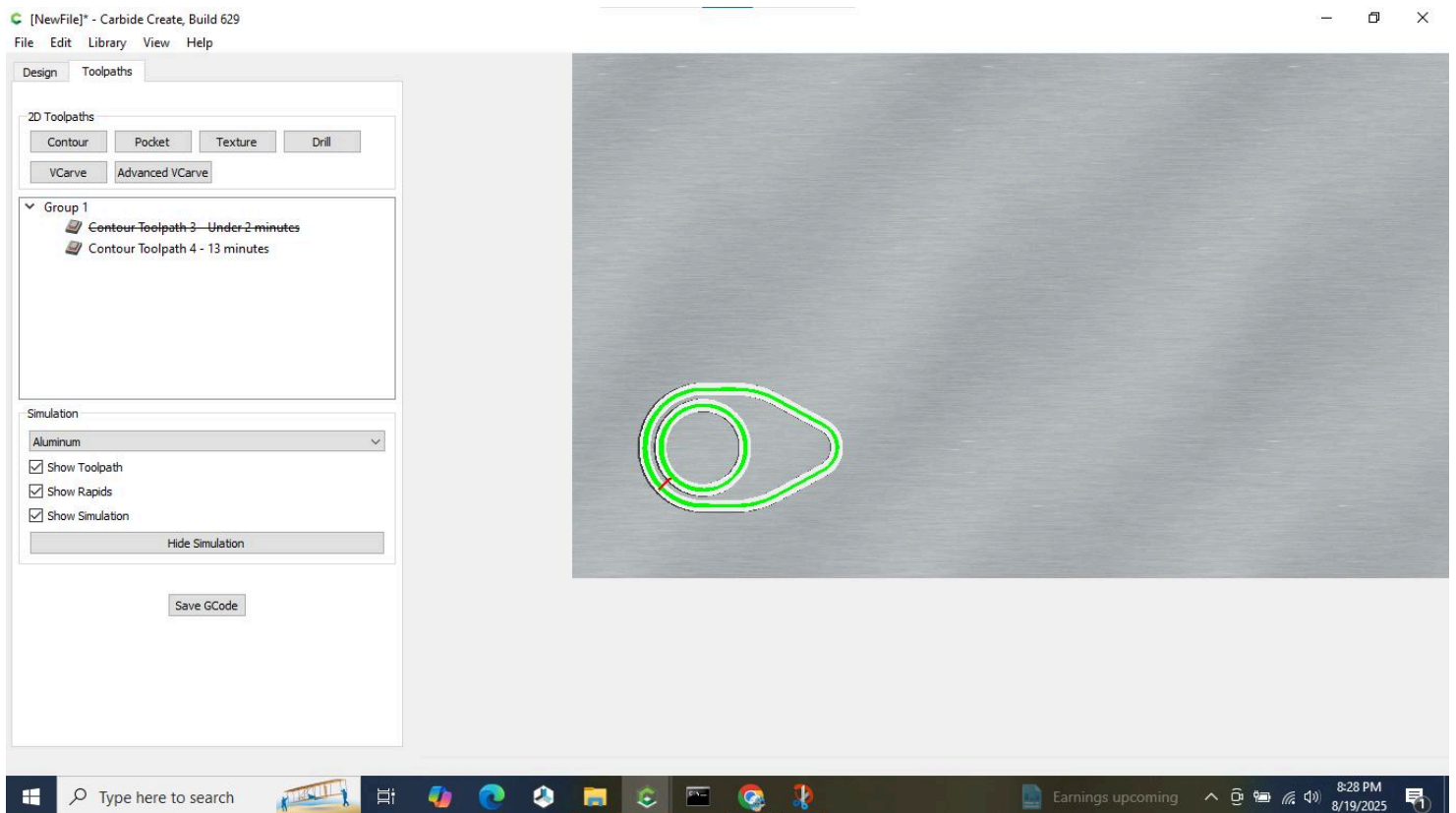
Save GCode



Go to contour and put the same values that you put for the hole toolpath, with one exception. This being the offset direction. This does matter depending on the amount of things you're cutting in the same toolpath (for example if you're just cutting the bearing hole and not the whole bearing piece, you can usually just use inside/left and it will work). But when you're using multiple outlines to cut at once (so for this one we are cutting the outline of the bearing cap itself & the bearing hole within the cap), you will most likely use outside/right. You can see in the image below that when we do outside/right, it cuts the bearing caps outline from outside to in (which is what we want if you want it to be precise) and the actual bearing hole from inside to out. Honestly I have no idea why it does that and why carbide create is set like that, but usually when you are cutting the outline of 2 things at once. you will be using outside/right.



Then go ahead and save that gcode the same way we did it before for the holes. (Make sure to do the whole M07/M09 thing if you're cutting aluminum). On top of that, you can see exactly what your cut will look like if you go to the show simulation button (which is right above the save gcode button). It will show you an outline of how it will cut (the red line being where it starts from and where it goes from there).



Bam, now you know how to make a gcode toolpath to cut something on carbide. But now how do you actually use that gcode to cut something? We're going to have to use fluidnc.

